



Product Name	KRABEVA 100mg/4mL & 400mg/16mL	Component	LEAFLET
Dimensions	Size: 630 (W) x 500 (H) mm Folding size: 70(W) x 50 (H) mm	Artwork Code	BF01839/01
Pharmacode	 12502	Country	MALAYSIA
Font Type	Times New Roman	Specification: 40gsm ITC news print paper	
Font Size	5.5 pt		
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The incidence of CNS bleeding in patients with untreated CNS metastases receiving bevacizumab has not been prospectively evaluated in randomised clinical studies. In an exploratory retrospective analysis of data from 13 completed randomised trials in patients with various tumour types, 3 patients out of 91 (3.3%) with brain metastases experienced CNS bleeding (all Grade 4) when treated with bevacizumab, compared to 1 case (Grade 5) out of 96 patients (1%) that were not exposed to bevacizumab. In two subsequent studies in patients with treated brain metastases (which included around 800 patients), one case of Grade 2 CNS haemorrhage was reported. Intracranial haemorrhage can occur in patients with relapsed glioblastoma. In study AVF3708g, CNS haemorrhage was reported in 2.4% (2/84) of patients in the Bevacizumab alone arm (Grade 1); and in 3.8% (3/79) of patients treated with Bevacizumab and irinotecan (Grades 1, 2 and 4). Across all Bevacizumab clinical trials, **mucoctaneous haemorrhage** were seen in up to 50% of patients treated with Bevacizumab. These were most commonly NCI-CTC Grade 1 epistaxis that lasted less than 5 minutes, resolved without medical intervention and did not require any change in the Bevacizumab treatment regimen. Clinical safety data suggest that the incidence of minor mucoctaneous haemorrhage (e.g. epistaxis) may be dose dependent. There have also been less common events of minor mucoctaneous haemorrhage in other locations, such as gingival bleeding or vaginal bleeding.

Hypertension (see section 2.4.1 Warnings and Precautions, General)
An increased incidence of hypertension (all grades) of up to 42.1% has been observed in patients treated with Bevacizumab compared with up to 14% in the comparator arm. In clinical trials across all indications the overall incidence of NCI-CTC Grade 3 and 4 hypertension in patients receiving Bevacizumab ranged from 0.4% to 17.9%. Grade 4 hypertension (hypertensive crisis) occurred in up to 1.0% of patients treated with Bevacizumab compared to up to 0.2% patients treated with the same chemotherapy alone. Hypertension was generally adequately controlled with oral anti-hypertensives such as angiotensin-converting enzyme inhibitors, diuretics and calcium-channel blockers. It rarely resulted in discontinuation of Bevacizumab treatment or hospitalisation. Very rare cases of hypertensive encephalopathy have been reported, some of which were fatal (see section 2.4.1 Warnings and Precautions, General). The risk of Bevacizumab associated hypertension did not correlate with the patients' baseline characteristics, underlying disease or concomitant therapy.

Posterior Reversible Encephalopathy Syndrome (see section 2.4.1 Warnings and Precautions, General)
Two confirmed cases (0.8%) of PRES have been reported in one clinical study. Symptoms usually resolve or improve within days, although some patients have experienced neurologic sequelae.

Thromboembolism - Arterial thromboembolism
An increased incidence of arterial thromboembolic events was observed in patients treated with Bevacizumab across indications including cerebrovascular accidents, myocardial infarction, transient ischemic attacks, and other arterial thromboembolic events. In clinical trials, the overall incidence ranged up to 5.9% in the Bevacizumab containing arms compared up to 1.7% in the chemotherapy control arms. Fatal outcome was reported in 0.8% of patients receiving Bevacizumab in combination with chemotherapy compared to 0.5% of patients receiving chemotherapy alone. Cerebrovascular accidents (including transient ischemic attacks) were reported in up to 2.7% of Bevacizumab treated patients versus 0.5% of patients in the control group; myocardial infarction was reported in 1.4% of Bevacizumab treated versus 0.7% of patients in the observed control group. In one clinical trial, AVF2192g, patients with metastatic colorectal cancer who were not candidates for treatment with irinotecan were included. In this trial arterial thromboembolic events were observed in 11% (11/100) of Bevacizumab patients compared to 5.8% (6/104) in the chemotherapy control group. In an uncontrolled clinical trial, AVF3708g, in patients with relapsed glioblastoma, arterial thromboembolic events were observed in 6.3% (5/79) of patients who received Bevacizumab in combination with irinotecan compared to 4.8% (4/84) of patients who received Bevacizumab alone.

- Venous thromboembolism (see section 2.4.1 Warnings and Precautions, General)
In clinical trials across indications, the overall incidence of venous thromboembolic events ranged from 2.8% to 17.3% in the Bevacizumab containing arms compared to 3.2% to 15.6% in the chemotherapy control arms. Venous thromboembolic events include deep vein thrombosis and pulmonary embolism. Grade 3-5 venous thromboembolic events have been reported in up to 7.8% of patients treated with chemotherapy plus bevacizumab compared with up to 4.9% in patients with chemotherapy alone. Patients who have experienced a venous thromboembolic event may be at higher risk for a recurrence if they receive Bevacizumab in combination with chemotherapy versus chemotherapy alone. From a clinical trial in patients with persistent, recurrent, or metastatic cervical cancer (study GOG-0240), grade 3-5 venous thromboembolic events have been reported in up to 10.6% of patients treated with chemotherapy and bevacizumab compared with up to 5.4% in patients with chemotherapy alone.

Congestive Heart Failure
As clinical trials with Bevacizumab, congestive heart failure (CHF) was observed in all cancer indications studied to date, but occurred predominantly in patients with metastatic breast cancer. In phase III studies (AVF2119g and E2100), in patients with metastatic breast cancer an increase of CHF Grade 3 or more with Bevacizumab was seen. CHF was reported in up to 3.5% of patients treated with Bevacizumab compared with up to 0.9% in the control arms. An increased incidence of post-operative wound healing complications occurring within 60 days of major surgery was observed if the patient was being treated with Bevacizumab at the time of surgery. The incidence varied between 10% (4/40) and 20% (3/15). Cases of serious wound healing complications have been reported during Bevacizumab use, some of which had a fatal outcome (see section 2.4.1 Warnings and Precautions, General). An increased incidence of CHF has been observed in a clinical trial of patients with diffuse large B-cell lymphoma when receiving bevacizumab with a cumulative doxorubicin dose greater than 300 mg/m². This phase III clinical trial compared rituximab/ cyclophosphamide/ epirubicin/ prednisone (R-CHOP) plus bevacizumab to R-CHOP without bevacizumab. While the incidence of CHF was, in both arms, above that previously observed for doxorubicin therapy, the rate was higher in the R-CHOP plus bevacizumab arm.

Wound Healing (see section 2.4.1 Warnings and Precautions, General)
As Bevacizumab may adversely impact wound healing, patients who had major surgery within the last 28 days prior to starting Bevacizumab treatment were excluded from participation in phase III trials. Across mCRC clinical trials there was no increased risk of post-operative bleeding or wound healing complications observed in patients who underwent major surgery between 28-60 days prior to starting Bevacizumab therapy. An increased incidence of post-operative bleeding or wound healing complications occurring within 60 days of major surgery was observed if the patient was being treated with Bevacizumab at the time of surgery. The incidence varied between 10% (4/40) and 20% (3/15). Cases of serious wound healing complications have been reported during Bevacizumab use, some of which had a fatal outcome (see section 2.4.1 Warnings and Precautions, General). In locally recurrent and metastatic breast cancer trials, Grade 3-5 wound healing complications were observed in up to 1.1% of patients receiving Bevacizumab compared with up to 0.9% of patients in the control arms. In a study of patients with relapsed glioblastoma (study AVF3708g), the incidence of postoperative wound healing complications (including cranial wound site wound dehiscence and cerebrospinal fluid leak) was 3.6% in patients treated with single-agent Bevacizumab and 1.3% in patients treated with Bevacizumab plus irinotecan. In clinical trials of ovarian cancer, Grade 3-5 wound healing complications were observed in up to 1.2% of patients in the bevacizumab arm versus 0.1% in the control arm.

Proteinuria (see section 2.4.1 Warnings and Precautions, General)
In clinical trials, proteinuria has been reported within the range of 0.7% to 54.7% of patients receiving Bevacizumab. Proteinuria ranged in severity from clinically asymptomatic, transient, trace proteinuria to nephrotic syndrome. Grade 3 proteinuria was reported in up to 8.1% of treated patients. Grade 4 proteinuria (nephrotic syndrome) was seen in up to 1.4% of treated patients. Patients with a history of hypertension may be at increased risk for the development of proteinuria when treated with Bevacizumab. There is evidence suggesting that Grade 1 proteinuria may be related to Bevacizumab dose. Testing for proteinuria is recommended prior to start of Bevacizumab therapy. In most clinical studies urine protein levels of $\geq$ 2g/24 hrs led to the holding of Bevacizumab until recovery to $<$ 2g/24hrs.

Hypersensitivity, infusion reactions (see section 2.4.1 Warnings and Precautions, General and section 2.6.2 Undesirable Effects, Postmarketing Experience)
In some clinical trials anaphylactic and anaphylatoxin-type reactions were reported more frequently in patients receiving Bevacizumab in combination with chemotherapies than with chemotherapy alone. The incidence of these reactions in some clinical trials of Bevacizumab is common (up to 5% in bevacizumab-treated patients).

Ovarian Failure/Fertility (see section 2.4.1 Warnings and Precautions, General and section 2.5.1 Use in Special Populations, Females and Males of Reproductive Potential)
The incidence of new cases of ovarian failure, defined as amenorrhea lasting 3 or more months, FSH level $\geq$ 30 mIU/mL and a negative serum B-HCG pregnancy test, has been evaluated. New cases of ovarian failure were reported more frequently in patients receiving bevacizumab. After discontinuation of bevacizumab treatment, ovarian function recovered in majority women. Long term effects of the treatment with bevacizumab on fertility are unknown.

Elderly Patients
In randomized clinical trials, age $>$ 65 years was associated with an increased risk of developing arterial thromboembolic events including cerebrovascular accidents,

transient ischemic attacks and myocardial infarction as compared to those aged $\leq$ 65 years when treated with Bevacizumab (see sections 2.4.1 Warnings and Precautions, General and section 2.6.1 Undesirable Effects, Clinical Trials, Thromboembolism). Other reactions with a higher frequency seen in patients over 65 were grade 3-4 leucopenia, thrombocytopenia; and all Grade neutropenia, diarrhoea, nausea, headache and fatigue. From a clinical trial in patients with metastatic colorectal cancer (study AVF2107g), no increase in the incidences of other reactions, including gastrointestinal perforation, wound healing complications, congestive heart failure and haemorrhage, was observed in elderly patients ($>$ 65 years) receiving Bevacizumab as compared to those aged $\leq$ 65 years treated with Bevacizumab.

Laboratory Abnormalities
Decreased neutrophil count, decreased white blood count and presence of urine protein may be associated with Bevacizumab treatment. Across clinical trials, the following Grade 3 and 4 laboratory abnormalities were seen with Bevacizumab (Table 2) based on spontaneous case reports and those in the control groups: hyperglycaemia, decreased haemoglobin, hypokalaemia, hyponatraemia, decreased white blood cell count, increased PT (prothrombin time), normalised ratio. Clinical trials have shown that transient increases in serum creatinine (ranging between 1.3-1.9 times baseline level), both with and without proteinuria, are associated with the use of Bevacizumab. The observed increase in serum creatinine was not associated with a higher incidence of clinical manifestations of renal impairment in patients treated with Bevacizumab.

2.6.2 Postmarketing Experience
The following adverse drug reactions have been identified from postmarketing experience with Bevacizumab (Table 2) based on spontaneous case reports and literature cases. Adverse drug reactions are listed according to system organ classes in MedDRA and the corresponding frequency category estimation for each adverse drug reaction is based on the following convention: very common ($\geq$ 1/10); common ($\geq$ 1/100 to $<$ 1/10); uncommon ($\geq$ 1/1,000 to $<$ 1/100); rare ($\geq$ 1/10,000 to $<$ 1/1,000); very rare ($<$ 1/10,000).

Table 2: Adverse Drug Reactions from Post marketing Experience

Adverse reactions	Frequency Category
Infections and Infestations	
Necrotising fasciitis ^{1,2}	Rare
Immune system disorders	
Hypersensitivity ^{2,4}	Unknown
Infusion reactions ³	Unknown
Nervous system disorders	
Hypertensive encephalopathy ^{1,4}	Very Rare
Posterior Reversible Encephalopathy Syndrome (PRES) ²	Rare
Vascular disorders	
Renal thrombotic microangiopathy, clinically manifested as proteinuria ^{1,4}	Unknown
Aneurysms and artery dissections	Unknown
Respiratory, thoracic and mediastinal disorders	
Nasal septum perforation	Unknown
Pulmonary hypertension	Unknown
Dysphonia	Common
Gastrointestinal disorders	
Gastrointestinal ulcer	Unknown
Hepatobiliary disorders	
Gallbladder perforation	Unknown
Musculoskeletal and connective tissue disorders	
Osteonecrosis of the Jaw (ONJ) ¹	Unknown
Osteonecrosis of sites other than the jaw ^{6,7}	Unknown
Congenital, familial and genetic disorders	
Foetal abnormalities ⁸	Unknown

- 1 Usually secondary to wound healing complications, gastrointestinal perforation or fistula formation
- 2 See section 2.4.1 Warnings and Precautions, General
- 3 The following are possible co-manifestations: dyspnoea/difficulty breathing, flushing/redness/rash, hypotension or hypertension, oxygen desaturation, chest pain, rigors and nausea/vomiting
- 4 See section 2.6.1 Undesirable Effects, Clinical Trials
- 5 Cases of ONJ observed in Bevacizumab-treated patients mainly in association with prior or concomitant use of bisphosphonates.
- 6 Cases observed in Bevacizumab-treated pediatric patients. See section 2.5.4 Use in special populations.
- 7 Pediatric use
- 8 Osteonecrosis observed in pediatric population in non-company clinical trials was identified through post-marketing surveillance and has therefore been added to the post-marketing section as neither CTC grade nor reporting rate were available from published data.
- 8 Cases have been observed in women treated with bevacizumab alone or in combination with known embryotoxic chemotherapeutics. See section 2.5.2 Use in Special Populations, Pregnancy.

Description of selected adverse drug reactions from postmarketing experience Eye disorders (reported from unapproved intravitreal use)
Infectious endophthalmitis (frequency not known); some cases leading to permanent blindness; one case reported extraocular extension of infection resulting in meningoencephalitis); Intraocular inflammation (some cases leading to permanent blindness; including a cluster of serious eye inflammation leading to blindness after compounding an anticancer chemotherapy product for intravenous administration) such as sterile endophthalmitis, uveitis, and vitritis; Retinal detachment (frequency not known); Retinal pigment epithelial tear (frequency not known); Intraocular pressure increased (frequency not known); Intraocular hemorrhage such as vitreous hemorrhage or retinal hemorrhage (frequency not known); Conjunctival hemorrhage (frequency not known).

Following variable and non-validated methods in compounding, storage, and handling of Bevacizumab, serious ocular adverse events (including infectious endophthalmitis and other ocular inflammatory conditions) affecting multiple patients have been reported.

Systemic Events (reported from unapproved intravitreal use)
Increased risk for haemorrhagic stroke; Increased risk for overall mortality. Increased risk of serious systemic metastatic events for Bevacizumab, most of which resulted in hospitalization (adjusted risk ratio 1.29; 95% CI: 1.01, 1.66) (Incidence 24.1%; comparator 19.0%).

2.7 Overdose
The highest dose tested in humans (20 mg/kg of body weight every 2 weeks, intravenous) was associated with severe migraine in several patients.

2.8 Interactions with Other Medicinal Products and Other Forms of Interactions

Effect of anti-neoplastic agents on bevacizumab pharmacokinetics

No clinically relevant pharmacokinetic interaction of co-administered chemotherapy on Bevacizumab pharmacokinetics has been observed based on the results of a population PK analysis. There was neither statistical significance nor clinically relevant difference in clearance of Bevacizumab in patients receiving Bevacizumab monotherapy compared to patients receiving Bevacizumab in combination with interferon-alfa 2a and other chemotherapies (i.e., 5-FU/LV, carboplatin/paclitaxel, capecitabine, doxorubicin or cisplatin/gemcitabine).

Effect of bevacizumab on the pharmacokinetics of other anti-neoplastic agents

No clinically relevant interaction of bevacizumab was observed on the pharmacokinetics of co-administered interferon-alfa 2a, or the chemotherapies irinotecan and 104 patients randomised to 5-FU/LV + Bevacizumab (5 mg/kg every 2 weeks). All of these findings were reversible upon discontinuation of bevacizumab by measurement of free and total platinum), and cisplatin. Conclusions on the impact of bevacizumab on gemcitabine pharmacokinetics cannot be drawn.

Combination of bevacizumab and sunitinib malate

In two clinical studies of metastatic renal cell carcinoma, microangiopathic haemolytic anaemia (MAHA) was reported in 7 of 19 patients treated with bevacizumab (10 mg/kg every two weeks) and sunitinib malate (50mg daily) combination. MAHA is a haemolytic disorder which can present with red cell fragmentation, anaemia, and thrombocytopenia. In addition, hypertension (including hypertensive crisis), elevated creatinine, and neurological symptoms were observed in some of these patients. All of these findings were reversible upon discontinuation of bevacizumab and sunitinib malate (see Hypertension, Proteinuria, PRES in section 2.4.1 Warnings and Precautions, General).

Radiotherapy

The safety and efficacy of concomitant administration of radiotherapy and Bevacizumab has not been established.

3. PHARMACOLOGICAL PROPERTIES AND EFFECTS

3.1 Pharmacodynamic Properties

3.1.1 Mechanism of Action

KRABEVA[®] (bevacizumab) is a recombinant humanised monoclonal antibody that selectively binds to and neutralises the biologic activity of human vascular endothelial growth factor (VEGF). Bevacizumab contains human framework regions with antigen binding regions of a humanised murine antibody that binds to VEGF. Bevacizumab is produced by recombinant DNA technology in a Chinese Hamster ovary mammalian cell expression system and is purified by a process that includes specific viral inactivation and removal steps. Bevacizumab consists of 214 amino acids and has a molecular weight of approximately 149,000 daltons. Bevacizumab inhibits the binding of VEGF to its receptors, Flt-1 and KDR, on the surface of endothelial cells. Neutralising the biologic activity of VEGF reduces the vascularisation of tumours, thereby inhibiting tumour growth. Administration of bevacizumab or its parental murine antibody to xenotransplant models of cancer in nude mice resulted in extensive antitumour activity in human cancers, including colon, breast, pancreas and prostate. Metastatic disease progression was inhibited and microvascular permeability was reduced.

3.1.2 Clinical/Efficacy Studies

Key data of KRABEVA[®]

Study MYL-14020-3001

A total of 671 patients were randomized with 337 patients to the KRABEVA[®] arm and 334 patients to the Avastin[®] arm. The primary efficacy endpoint of this Phase III study in NSCLC was the ORR as assessed by an independent review during the first 18 Weeks, according to RECIST 1.1. At Week 18, 217 (64.4%) patients in the KRABEVA[®] arm and 225 (67.4%) in the Avastin[®] arm had their treatment ongoing. The primary efficacy endpoint was based on the ratio of ORRs. The equivalence of both treatments was defined as the two-sided 90% CI for the ratio of ORRs at Week 18 being entirely within the equivalence range of 0.73, 1.36. The ratio of ORR (KRABEVA[®]: Avastin[®]) between the treatment groups was 0.96 (90% CI 0.83, 1.12). As this CI was entirely within the pre-defined equivalence margin, therapeutic equivalence of KRABEVA[®] and Avastin[®] was statistically confirmed.

Immunogenicity Results

A total of 12 (3.6%) and 16 patients (5.0%) were positive for ADA at baseline in the KRABEVA[®] and Avastin[®] arms, respectively. The incidence of baseline ADA positivity in a small proportion of patients has been observed in other bevacizumab studies and may be due to potential cross-reactivity with pre-existing antibodies and highly sensitive nature of the assay. Post-baseline, the number of ADA positive patients were low, declined over time and were comparable between both treatment arms. The ADA titers were found to be comparable in the 2 arms.

Study BM100-CC-03-01

Efficacy of KRABEVA[®] and Avastin (both in combination with XELOX chemotherapy) was compared in terms of best overall response rate (BORR) over 18 weeks in a Phase III study in patients with mCRC. BORR was determined in the PP as well as the ITT populations where objective response was defined as a CR or PR. PR was observed in 25 patients in the KRABEVA[®] arm and 32 patients in the Avastin arm. The ORR observed in the ITT population was 38.2% in the KRABEVA[®] arm and 48.53% in the Avastin[®] arm. No statistically significant difference was observed between the two arms (p = 0.2258). The ORR observed in this study is similar to historically observed ORR of 36.6% when bevacizumab is used in combination with chemotherapy.

The PFS rate was similar in both the arms at 18 weeks (61.76% in the KRABEVA[®] and 60.29% in the Avastin arm). No statistically significant difference between the two arms was observed for PFS rate (p = 0.8604).

Due to the prominent effect of bevacizumab on disease stabilization, clinical benefit to the patients based on DCR was analyzed and compared between the two arms. DCR is defined as the sum of CR, PR and SD based on the best overall response evaluated as per the RECIST 1.1. The DCR was 91.18% in the KRABEVA[®] arm and 88.24% in the Avastin arm in the ITT population. The analysis of the DCR revealed no notable differences between both the arms for the proportion of patients with disease control (p=0.5399).

All studies performed in every single indication as approved are referring to the studies conducted by AVASTIN.

Metastatic Colorectal Cancer (mCRC)

The safety and efficacy of the recommended dose of Bevacizumab (5 mg/kg of body weight every two weeks) in metastatic carcinoma of the colon or rectum were studied in three randomized, active-controlled clinical trials in combination with fluoropyrimidine-based first-line chemotherapy. Bevacizumab was combined with two chemotherapy regimens:

AVF2107g: A weekly schedule of irinotecan/bolus 5-fluorouracil/leucovorin (IFL regimen) for a total of 4 weeks of each 6-week cycle.

AVF780g: In combination with bolus 5-fluorouracil/leucovorin (5-FU/LV) for a total of 6 weeks of each 8-week cycle (Roswell Park regimen).

AVF2102g: In combination with bolus 5-fluorouracil/leucovorin (5-FU/LV) for a total of 6 weeks of each 8-week cycle (Roswell Park regimen) in patients who were not optimal candidates for first-line irinotecan treatment.

Three additional studies with Bevacizumab have been conducted in mCRC patients: first-line (N016966), second-line with no previous Bevacizumab treatment (E3200), and second-line with previous Bevacizumab treatment following disease progression in first-line (ML18147). In these studies, Bevacizumab was administered at the following dosing regimens, in combination with FOLFOX-4 (5FU/LV/Oxaliplatin), XELOX (Capecitabine/Oxaliplatin) and fluoropyrimidine/irinotecan and fluoropyrimidine/oxaliplatin:

- **N016966:** Bevacizumab 7.5 mg/kg of body weight every 3 weeks in combination with oral capecitabine and intravenous oxaliplatin (XELOX) or Bevacizumab 5mg/kg every 2 weeks in combination with leucovorin plus 5-fluorouracil bolus, followed by 5-fluorouracil infusion, with intravenous oxaliplatin (FOLFOX-4)
- **E3200:** Bevacizumab 10 mg/kg of body weight every 2 weeks in combination with leucovorin and 5-fluorouracil bolus, followed by 5-fluorouracil infusion, with intravenous oxaliplatin (FOLFOX-4) in Bevacizumab-naïve patients.
- **ML18147:** Bevacizumab 5.0 mg/kg of body weight every 2 weeks or Bevacizumab 7.5 mg/kg of body weight every 3 weeks in combination with fluoropyrimidine/irinotecan or fluoropyrimidine/oxaliplatin in patients with disease progression following first-line treatment with Bevacizumab. Use of irinotecan- or oxaliplatin-containing regimen was switched depending on first-line use of either oxaliplatin or irinotecan.

AVF2107g:

This was a phase III randomised, double-blind, active-controlled clinical trial evaluating Bevacizumab in combination with IFL as first-line treatment for metastatic carcinoma of the colon or rectum. Eight hundred and thirteen patients were randomised to receive IFL + placebo (Arm 1) or IFL + Bevacizumab (5 mg/kg every 2 weeks; Arm 2). A third group of 110 patients received bolus 5-FU/LV + Bevacizumab (Arm 3). Enrolment in Arm 3 was discontinued, as pre-specified, once safety of Bevacizumab with the IFL regimen was established and considered acceptable.

The primary efficacy parameter of the trial was overall survival. The addition of Bevacizumab to IFL resulted in a statistically significant increase in overall survival, progression-free survival and overall response rate (see Table 3 for details). The clinical benefit of Bevacizumab, as measured by survival, was seen in all pre-specified patient subgroups, including those defined by age, sex, performance status, and location of primary tumour, number of organs involved, and duration of metastatic disease.

Table 3 Efficacy Results for Study AVF2107g

	AVF2107g	
	Arm 1 IFL + Placebo	Arm 2 IFL + Bevacizumab ^a
Number of Patients	411	402
Overall Survival		
Median (months)	15.6	20.3
95% confidence interval	14.29 – 16.99	18.46 – 24.18
Hazard ratio ^b	0.660 (p-value = 0.00004)	
Secondary endpoint: Progression-Free Survival		
Median (months)	6.2	10.6
Hazard ratio	0.54 (p-value < 0.00001)	
Overall Response Rate	34.8%	44.8% (p-value = 0.0036)

^a 5 mg/kg every 2 weeks

^b Relative to control arm

Among the 110 patients randomised to Arm 3 (5-FU/LV + Bevacizumab) prior to discontinuation of this arm, the median overall survival was 18.3 months, and the median progression free survival was 8.8 months.

AVF2192g

This was a phase II randomised, double-blind, active-controlled clinical trial investigating Bevacizumab in combination with 5-FU/leucovorin as first-line treatment for metastatic colorectal cancer in patients who were not optimal candidates for first-line irinotecan treatment. One hundred and five patients were randomised to 5-FU/LV + placebo arm and 104 patients randomised to 5-FU/LV + Bevacizumab (5 mg/kg every 2 weeks). All treatments were continued until disease progression. The addition of Bevacizumab 5 mg/kg every two weeks to 5-FU/LV resulted in higher objective response rates, significantly longer progression-free survival, and a trend in longer survival as, compared with 5-FU/LV chemotherapy alone.

N016966

This was a phase III randomised, double-blind (for bevacizumab), clinical trial investigating Bevacizumab 7.5 mg/kg in combination with oral capecitabine and i.v. oxaliplatin (XELOX), administered on a 3-weekly schedule; or Bevacizumab 5 mg/kg in combination with leucovorin and 5-fluorouracil bolus, followed by 5-fluorouracil infusion, with i.v. oxaliplatin (FOLFOX-4), administered on a 2-weekly schedule. The study contained two parts: an initial unblinded 2-arm part (Part I) in which patients were randomized to two different treatment groups (XELOX and FOLFOX-4) and a subsequent 2 x 2 factorial 4-arm part (Part II) in which patients were randomized to four treatment groups (XELOX + placebo, FOLFOX-4 + placebo, XELOX + Bevacizumab, FOLFOX-4 + Bevacizumab). In Part II, treatment assignment was double-blind with respect to Bevacizumab.

Approximately 350 patients were randomized into each of the 4 study arms in the Part II of the trial.

Table 4 Treatment Regimens in Study N016966 (mCRC)

Treatment	Starting Dose	Schedule
FOLFOX-4 or FOLFOX-4 + Bevacizumab	Oxaliplatin 85 mg/m ² IV 2 h Leucovorin 200 mg/m ² IV 2 h 5-Fluorouracil 400 mg/m ² IV bolus, 600 mg/m ² IV 22 h	Oxaliplatin on Day 1 Leucovorin on Day 1 and 2 5-Fluorouracil IV bolus/infusion, each on Days 1 and 2 Day 1, prior to FOLFOX-4, every 2 weeks
XELOX or XELOX + Bevacizumab	Placebo or Bevacizumab 5 mg/kg IV 30-90 min Oxaliplatin 130 mg/m ² IV 2 h Capecitabine 1300 mg/m ² oral bid	Oxaliplatin on Day 1 Capecitabine oral bid for 2 weeks (followed by 1 week off treatment) Day 1, prior to XELOX, q 3 weeks
5-Fluorouracil: IV bolus injection immediately after leucovorin	Placebo or Bevacizumab 7.5 mg/kg IV 30-90 min	Day 1, prior to XELOX, q 3 weeks

The primary efficacy parameter of the trial was the duration of progression-free survival. In this study, there were two primary objectives: to show that XELOX was non-inferior to FOLFOX-4 and to show that Bevacizumab in combination with

FOLFOX-4 or XELOX chemotherapy was superior to chemotherapy alone. Both co-primary objectives were met:

i) Non-inferiority of the XELOX-containing arms compared with the FOLFOX-4-containing arms in the overall comparison was demonstrated in terms of progression-free survival and overall survival in the eligible per-protocol population.

ii) Superiority of the Bevacizumab-containing arms versus the chemotherapy alone arms in the overall comparison was demonstrated in terms of progression-free survival in the ITT population (Table 5).

Secondary PFS analyses, based on Independent Review Committee (IRC) - and 'on treatment'- based response assessments, confirmed the significantly superior clinical benefit for patients treated with Bevacizumab (subgroup analyses shown in Table 5), consistent with the statistically significant benefit observed in the pooled analysis.

Table 5 Key efficacy results for the superiority analysis (ITT population, Study N016966)

Endpoint (months)	FOLFOX-4 or XELOX + Placebo (n=701)	FOLFOX-4 or XELOX + Bevacizumab (n=699)	P Value
Primary endpoint			
Median PFS**	8.0	9.4	0.0023
Hazard ratio (97.5% CI) ^b	0.83 (0.72–0.95)		
Secondary endpoints			
Median PFS (on treatment)**	7.9	10.4	<0.0001
Hazard ratio (97.5% CI)	0.63 (0.52-0.75)		
Median PFS (Indep. review)**	8.5	11.0	<0.0001
Hazard ratio (97.5% CI)	0.70 (0.58-0.83)		
Overall response rate (Invest. Assessment)**	49.2%	46.5%	
Overall response rate (Indep. Review)**	37.5%	37.5%	
Median overall survival ^a	19.9	21.2	0.0769
Hazard ratio (97.5% CI)	0.89 (0.76-1.03)		

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IRF-assessed DOR

Median in months

95% CI

6-month DOR (%)

Stratified by geographic region (Asia excluding Japan vs. rest of world), macrovascular invasion and/or extrahepatic spread (presence vs. absence), and baseline AFP (<400 vs. ≥400 ng/mL)

Based on two-sided stratified log-rank test

Based on two-sided Cochran-Mantel-Haenszel test

PFS=progression-free survival; RECIST=Response Evaluation Criteria in Solid Tumors v1.1; HCC mRECIST= Modified RECIST Assessment for Hepatocellular Carcinoma ; CI=confidence interval; ORR=objective response rate; DOR=duration of response; OS=overall survival; NE=not estimable; N/A=not applicable

Key efficacy endpoints	Tecentriq + Avasitin	Sorafenib
OS	n=336	n=165
No. of deaths (%)	180 (53.6%)	100 (60.6%)
Median time to event (months)	19.2	13.4
95% CI	(17.0, 23.7)	(11.4, 16.9)
Stratified hazard ratio [†] (95% CI)	0.66 (0.52, 0.85)	
IRF-assessed ORR, RECIST I.I	n=326	n=159
No. of confirmed responders (%) [*]	97 (29.8%)	18 (11.3%)
95% CI	(24.8, 35.0)	(6.9, 17.3)
IRF-assessed DOR, RECIST I.I	n=97	n=18
Median in months	18.1	14.9
95% CI	(14.6, NE)	(4.9, 17.0)
[†] Stratified by geographic region (Asia excluding Japan vs rest of world), macrovascular invasion and/or extrahepatic spread (presence vs. absence), and baseline AFP (<400 vs. ≥400 ng/mL)		
[*] No. of complete responses (%): 25 (7.7%) in the Tecentriq + Avasitin arm and 1 (0.6%) in the sorafenib arm		
PFS=progression-free survival; RECIST=Response Evaluation Criteria in Solid Tumors v1.1; CI=confidence interval; ORR=objective response rate; DOR=duration of response; OS=overall survival; NE=not estimable		

Figure 1: Kaplan-Meier curve for Overall Survival (IMbrave150 Updated Analysis)

Figure 2: Kaplan-Meier Plot for Progression-Free Survival per RECIST v1.1 (IMbrave150 Primary Analysis)

720 mm

510 mm

5 mm

5 mm

Text Free Area

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3.2.4 Elimination

The pharmacokinetics of bevacizumab are linear at doses ranging from 1.5 to 10 mg/kg/wk.

The value for clearance is, on average, equal to 0.188 and 0.220 L/day for female and male patients, respectively. After correcting for body weight, male patients had a higher bevacizumab clearance (+17%) than females. According to the two-compartmental model, the elimination half-life is 18 days for a typical female patient and 20 days for a typical male patient.

3.2.5 Pharmacokinetics in Special Populations

The population pharmacokinetics of bevacizumab were analysed to evaluate the effects of demographic characteristics. In adults, the results showed no significant difference in the pharmacokinetics of bevacizumab in relation to age.

Pediatric population: The pharmacokinetics of bevacizumab were evaluated in 152 patients (7 months to 21 years; 5.9 to 125kg) across 4 clinical studies using a population pharmacokinetic model. The pharmacokinetic results show that the clearance and the volume of distribution of bevacizumab were comparable between pediatric and adult patients when normalized by body-weight. Age was not associated with the pharmacokinetics of bevacizumab when body-weight was taken into account.

Renal impairment: No studies have been conducted to investigate the pharmacokinetics of bevacizumab in renally impaired patients since the kidneys are not a major organ for bevacizumab metabolism or excretion.

Hepatic impairment: No studies have been conducted to investigate the pharmacokinetics of bevacizumab in patients with hepatic impairment since the liver is not a major organ for bevacizumab metabolism or excretion.

3.3 Nonclinical Safety

3.3.1 Carcinogenicity

Studies have not been performed to evaluate the carcinogenic potential of Bevacizumab.

3.3.2 Genotoxicity

Studies have not been performed to evaluate the mutagenic potential of Bevacizumab.

3.3.3 Impairment of Fertility

No specific studies in animals have been performed to evaluate the effect of Bevacizumab on fertility. No adverse effect on male reproductive organs was observed in repeat dose toxicity studies in cynomolgus monkeys.

Inhibition of ovarian function was characterised by decreases in ovarian and/or uterine weight and the number of corpora lutea, a reduction in endometrial proliferation and an inhibition of follicular maturation in cynomolgus monkeys treated with Bevacizumab for 13 or 26 weeks. The doses associated with this effect were ≥ 4 times the human therapeutic dose or ≥ 2-fold above the expected human exposure based on average serum concentrations in female monkeys. In rabbits, administration of 50 mg/kg of bevacizumab resulted in a significant decrease in ovarian weight and number of corpora lutea. The results in both monkeys and rabbits were reversible upon cessation of treatment. The inhibition of angiogenesis following administration of bevacizumab is likely to result in an adverse effect on female fertility.

3.3.4 Reproductive Toxicity

Bevacizumab has been shown to be embryotoxic and teratogenic when administered to rabbits. Observed effects included decreases in maternal and foetal body weights, an increased number of foetal resorptions and an increased incidence of specific gross and skeletal foetal alterations. Adverse foetal outcomes were observed at all tested doses of 10–100 mg/kg.

Information on foetal malformations observed in the post marketing setting are provided in section 2.5.1 Use in Special Populations, Pregnancy and 2.6.2 Undesirable Effects, Postmarketing Experience.

3.3.5 Other

Physical development:

In studies of up to 26 weeks duration in cynomolgus monkeys, Bevacizumab was associated with physical dysplasia. Physical dysplasia was characterised primarily by thickened growth plate cartilage, subchondral bony plate formation and inhibition of vascular invasion of the growth plate. This effect occurred at doses ≥ 0.8 times the human therapeutic dose and exposure levels slightly below the expected human clinical exposure, based on average serum concentrations. It should be noted, however, that physical dysplasia occurred only in actively growing animals with open growth plates.

Wound healing:

In rabbits, the effects of bevacizumab on circular wound healing were studied. Wound reepithelialisation was delayed in rabbits following five doses of bevacizumab, ranging from 2-50 mg/kg, over a 2-week period. A trend toward a dose-dependent relationship was observed. The magnitude of effect on wound healing was similar to that observed with corticosteroid administration. Upon treatment cessation with either 2 or 10 mg/kg bevacizumab, the wounds closed completely. The lower dose of 2 mg/kg was approximately equivalent to the proposed clinical dose. A more sensitive linear wound healing model was also studied in rabbits. Three doses of bevacizumab ranging from 0.5-2 mg/kg dose dependently and significantly decreased the tensile strength of the wounds, consistently with delayed wound healing. The low dose of 0.5 mg/kg was 5-fold below the proposed clinical dose.

As effects on wound healing were observed in rabbits at doses below the proposed clinical dose, the capacity for bevacizumab to adversely impact wound healing in human should be considered.

In cynomolgus monkeys, the effects of bevacizumab on the healing of a linear incision were highly variable and no dose-response relationship was evident.

Renal function:

In normal cynomolgus monkeys, bevacizumab had no measurable effect on renal function treated once or twice weekly for up to 26 weeks, and did not accumulate in the kidney of rabbits following two doses up to 100 mg/kg (approximately 80-folds the proposed clinical dose).

Investigative toxicity studies in rabbits, using the models of renal dysfunction, showed that bevacizumab did not exacerbate renal glomerular injury induced by bovine serum albumin or renal tubular damage induced by cisplatin.

Albumin:

In male cynomolgus monkeys, Bevacizumab administered at doses of 10 mg/kg twice weekly or 50 mg/kg once weekly for 26 weeks was associated with a statistically significant decrease in albumin and albumin to globulin ratio and increase in globulin. These effects were reversible upon cessation of exposure. As the parameters remained within the normal reference range of values for these endpoints, these changes were not considered as clinically significant.

Hypertension:

At doses up to 50 mg/kg twice weekly in cynomolgus monkeys, Bevacizumab showed no effects on blood pressure.

Haemostasis:

Non-clinical toxicology studies of up to 26 weeks duration in cynomolgus monkeys did not find changes in hematology or coagulation parameters including platelet counts, prothrombin and activated partial thromboplastin time. A model of hemostasis in rabbits, used to investigate the effect of bevacizumab on thrombus formation, did not show alteration in the rate of clot formation or any other hematological parameters compared to treatment with Bevacizumab vehicle.

4. PHARMACEUTICAL PARTICULARS

4.1 Storage

Shelf Life

Please refer to carton/label.

Storage

Store vials in a refrigerator at 2°C–8°C. Keep out of reach of children.

Keep vial in the outer carton in order to protect from light.

DO NOT FREEZE. DO NOT SHAKE.

KRABEVA[®] does not contain any antimicrobial preservative; therefore, care must be taken to ensure the sterility of the prepared solution.

KRABEVA[®] is stable and compatible with sterile saline solution (0.9% sodium chloride solution) up to 70 days at 2°C – 8°C and a period of up to 15 days at 23° - 27°C. From a microbiological point of view, the product should be used immediately.

4.2 Special Instructions for Use, Handling and Disposal

KRABEVA[®] infusions should not be administered or mixed with dextrose or glucose solutions (see "Incompatibilities" below)

Do not administer as an intravenous push or bolus.

KRABEVA[®] should be prepared by a healthcare professional using aseptic technique. Withdraw the necessary amount of KRABEVA[®] and dilute to the required administration volume with 0.9% sodium chloride solution. The concentration of the final KRABEVA[®] solution should be kept within the range of 1.4–16.5 mg/mL.

Discard any unused portion left in a vial, as the product contains no preservatives. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration.

KRABEVA[®] is not formulated for intravitreal use.

Incompatibilities

No incompatibilities between KRABEVA[®] and polyvinyl chloride or polyolefin bags have been observed. Dilute KRABEVA[®] with using normal saline only. Do not prepare or administer in Dextrose or glucose solutions.

Disposal of unused/expired medicines

The release of pharmaceuticals in the environment should be minimised. Medicines should not be disposed of via wastewater and disposal through household waste should be avoided. Use established "collection systems", if available in your location.

PACKAGING INFORMATION

KRABEVA[®] 100 mg/4 mL – Pack of 1 vial (4 mL):

KRABEVA[®] 100 mg (25mg/mL) is filled in USP type 1 glass vial, plugged with a stopper (Chlorobutyl rubber) and sealed with flip-off seal of aluminium and plastic (polypropylene)

Manufacturer and Batch Releaser

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